

This week we have two short programs to consider. For each of them, please submit a copy of the source code and a screenshot of the program running.

1. *Factorial*

Use **scanf** (or **scanf_s**) to request an integer number from the user. Write a program using a **for** loop to print the factorial of the number. Your program should work properly for any number between 1 and 20 inclusive. Please submit your code and a screenshot of it working.

2. *Musical notes*

Western music (that is, the musical tradition originating in Western Europe) uses a twelve-tone equal temperament tuning system. Within this system, the notes have frequencies given by:

$$f_k = f_0 \alpha^k$$

where f_k is the frequency of the k^{th} note (k is an integer), f_0 and α are constants. We define α to provide twelve equally spaced (on a logarithmic scale) notes per octave:

$$\alpha = \exp \frac{\ln(2)}{12} = \sqrt[12]{2}$$

The other constant, f_0 , is concert A, typically 440 Hz in modern music. A full size piano keyboard covers 88 notes from $k = -48$ to $k = +39$. Your task is to write a program that uses a **for** statement to list the frequency of each of the 88 notes that are found on a piano keyboard. You may find the **exp**, **log**, and **pow** functions from the **math.h** header file handy. The **log** function computes the natural log of a number. The **exp** function returns the base e exponential function. The **pow** function computes a^b . Examples appear below:

```
x = log (a);
x = exp (a);
x = pow (a,b);
```

Your program should print k and the frequency of the note in two columns. Please use the following **printf** statement in your code:

```
printf("%3i, %9.2f\n", k, note);
```

The **%3i** format specifier prints an integer with a minimum width of three characters (if less than three characters are required, spaces are printed to the left of the number to make three characters; a negative sign counts as a character). The **%9.2f** format specifier prints a floating point number with a minimum width of nine characters, including two digits after the decimal point (the decimal point also counts as a character).

Because of the large exponents, α should be typed as a **double** to ensure the accuracy of your results. The other real numbers can be **floats**. The first lines of your program's output should appear as shown below:

key#	frequency [Hz]
-48,	27.50
-47,	29.14
-46,	30.87
-45,	32.70
-44,	34.65